

SSEE-V3.6 in the Strong-Gravity Regime: Disformal Geodesics, MIRA Emergence, and Vainshtein Screening

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Abstract

We extend the SSEE-V3.6 effective field theory framework to the strong-gravity regime, where the disformal coupling between the dark-energy scalar ϕ and dark matter generates deviations from general relativity (GR) that are algebraically fixed by the golden ratio φ and π . Starting from the k -essence action of Paper 7, we derive the quasi-static scalar field equation and show that the gradient closure $\nabla\phi = 2\text{KAL}_0 \beta_c M_{\text{Pl}} \nabla\Phi_N$ links the scalar profile to the Newtonian potential through the structural coupling $\beta_c \equiv \mathcal{A} = (3\varphi + \pi)/2 \approx 3.998$. From the disformal null geodesic, we demonstrate that the total photon deflection angle is amplified by the factor $\sqrt{\beta_c}$ relative to the GR prediction for the same baryonic mass. This yields an algebraic prediction for the Einstein radius ratio: $\theta_{\text{E}}^{\text{SSEE}}/\theta_{\text{E}}^{\text{GR}} = \sqrt{\beta_c} \approx \mathcal{M}$, where $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$ is the same structural constant that maps the acoustic horizon in the CMB sector (Paper 3); the 0.03% gap $\sqrt{\beta_c}/\mathcal{M} = 1.00030$ is a near-coincidence, not an algebraic identity (see Section 5). Solar-system gravitational tests are automatically satisfied because the disformal coupling selects dark matter as the sole source of ϕ : in the solar system $\rho_{\text{DM}} \approx 0$, so ϕ is unsourced locally and the disformal modification to the metric is negligible. We derive the two-sector lensing prediction: on scales $k < k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$ both CDM and ϕ -DM contribute and the Einstein radius matches the Λ CDM prediction; on scales $k > k_{\text{fs}}$ only the CDM sector contributes and the effective lensing mass is suppressed by $\sim \mathcal{M}$. These predictions are falsifiable with current HST and JWST strong-lensing surveys.

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1 Introduction

Papers 1–7 of the SSEE series established the cosmological background (Almeida Vallejo, 2025a), Bayesian validation (Almeida Vallejo, 2025b), CMB confrontation (Almeida Vallejo, 2025c), algebraic CMB derivation (Almeida Vallejo, 2025d), Israel–Stewart perturbation theory (Almeida Vallejo, 2026a), two-sector ϕ -DM phenomenology (Almeida Vallejo, 2026b), and the canonical EFT formulation (Almeida Vallejo, 2026c). All observable predictions in those papers operate within the linear or quasi-linear regime.

The disformal coupling that defines SSEE has observable consequences in the strong-gravity regime that have not yet been examined. In coupled dark-energy models with disformal metrics, two qualitatively new effects arise beyond the linearised regime:

1. **Enhanced deflection angle** — photons propagating through the modified gravitational potential of disformally-coupled dark matter experience a different total deflection relative to pure GR.
2. **Vainshtein suppression** — the non-linear scalar kinetic term in the SSEE action screens the fifth force on solar-system and stellar scales, restoring GR locally.

In this paper we derive both effects analytically, obtain numerical predictions using the SSEE algebraic constants, and identify the structural constant $\sqrt{\beta_c} \approx \mathcal{M}$ (to 0.03%) as the lensing amplification factor, where $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$. This is the same constant that appears in the CMB horizon mapping (Paper 3) and in the two-sector mass ratio (Paper 6):

$$\mathcal{M} = \frac{3\varphi + \pi}{4} = \frac{\Omega_{m,\text{CMB}}}{\Omega_{m,\text{dyn}}} = \frac{\beta_c}{2} = \frac{\mathcal{A}}{2}, \quad (1)$$

where $\beta_c \equiv \mathcal{A} = (3\varphi + \pi)/2$ is the disformal coupling constant fixed in Paper 7. That $\mathcal{M}$ appears in both the background (CMB horizon) and the perturbation (lensing) sectors is a non-trivial internal consistency check of the algebraic structure.

The paper is organised as follows. Section 2 recalls the relevant EFT ingredients from Paper 7. Section 3 derives the quasi-static scalar EOM and gradient closure in the non-relativistic limit. Section 4 constructs the disformal null geodesic and derives the photon deflection angle. Section 5 shows algebraically how $\mathcal{M}$ emerges as the lensing amplification factor. Section 6 derives the Vainshtein radius and the screening suppression. Section 7 translates the results into strong-lensing observables and identifies JWST/HST falsification windows. Section 8 collects the pre-committed falsifiable predictions. Section 9 discusses the two-sector scale dependence and caveats. Section 10 concludes.

2 EFT Foundation: Action and Disformal Coupling

We adopt the SSEE canonical action from Paper 7 (Almeida Vallejo, 2026c):

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + P(X, \phi) \right] + S_{\text{DM}}[\tilde{g}_{\mu\nu}; \psi_{\text{DM}}] + S_b[g_{\mu\nu}; \psi_b], \quad (2)$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ is the canonical kinetic invariant, and the k -essence kernel is

$$P(X, \phi) = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4} - V_0 e^{-\lambda\phi/M_{\text{Pl}}}. \quad (3)$$

Baryons (ψ_b) couple to the physical metric $g_{\mu\nu}$; dark matter (ψ_{DM}) couples to the disformal metric

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{2}{M^4} \partial_\mu \phi \partial_\nu \phi. \quad (4)$$

The algebraic constants fixed by φ and π (Paper 7) are:

$$\text{KAL}_0 = \beta + \pi = \frac{\varphi + \pi}{2} + \pi \approx 5.521, \quad (5)$$

$$\beta_c \equiv \mathcal{A} = \frac{3\varphi + \pi}{2} \approx 3.998, \quad (6)$$

$$\mathcal{M} = \frac{\beta_c}{2} = \frac{3\varphi + \pi}{4} \approx 1.9989, \quad (7)$$

$$M \approx M_{\text{Pl}} \quad (\text{free EFT parameter; we adopt } M = M_{\text{Pl}} \text{ as reference}). \quad (8)$$

The parameter β_c encodes the strength of the disformal coupling; it appears both in the coupling term of (4) (through M) and in the conformal-sector coupling Q^ν of Paper 7.

For the purposes of this paper, we work in the quasi-static, non-relativistic limit ($v \ll c$, $\Phi_N \ll 1$), which is appropriate for galaxy and cluster scales where $k \gg aH$.

3 Scalar Field Equation and Gradient Closure

3.1 Quasi-static limit

Varying the action (2) with respect to ϕ gives the scalar equation of motion. In the quasi-static approximation ($|\dot{\phi}| \ll |\nabla\phi|$, $|\ddot{\phi}| \ll |\nabla^2\phi|$) and at leading order in the kinetic expansion, the non-linear kinetic term X^2/M^4 is subdominant on sub-Hubble scales. The equation reduces to

$$\frac{1}{\text{KAL}_0} \nabla^2 \phi = \frac{\beta_c}{M_{\text{Pl}}} \rho_{\text{DM}}, \quad (9)$$

where the right-hand side arises from the disformal coupling of dark matter to $\tilde{g}_{\mu\nu}$: dark matter acts as an effective source for ϕ with coupling strength β_c/M_{Pl} .

The Newtonian gravitational potential Φ_N satisfies the standard Poisson equation

$$\nabla^2 \Phi_N = \frac{\rho_{\text{DM}} + \rho_b}{2M_{\text{Pl}}^2}, \quad (10)$$

where we use units $8\pi G = 1/M_{\text{Pl}}^2$.

3.2 Gradient closure

In the dark-matter-dominated regime ($\rho_{\text{DM}} \gg \rho_b$), combining (9) and (10) yields the gradient closure relation

$$\boxed{\nabla\phi = 2\text{KAL}_0 \beta_c M_{\text{Pl}} \nabla\Phi_N.} \quad (11)$$

This is the central structural relation of Paper 8: the scalar gradient tracks the Newtonian potential gradient with an algebraic coefficient $2\text{KAL}_0 \beta_c \approx 2 \times 5.521 \times 3.998 \approx 44.15$.

Equation (11) implies that wherever there is a gravitational potential well, the scalar field ϕ builds up a corresponding profile. This is not an assumption but a consequence of the disformal coupling structure in the quasi-static limit. We note that this derivation assumes $\rho_{\text{DM}} \gg \rho_b$; in systems with significant baryonic fraction, equation (9) should be replaced by the full source including baryons, with a smaller effective coupling.

3.3 Scalar force on dark matter

The effective force on a dark matter particle in this scalar background is

$$F_\phi = -\frac{\beta_c}{M_{\text{Pl}}} \nabla \phi = -2\text{KAL}_0 \beta_c^2 \nabla \Phi_N. \quad (12)$$

The total force on a dark matter test particle is $F_{\text{tot}} = F_N + F_\phi = -(1 + 2\text{KAL}_0 \beta_c^2) \nabla \Phi_N$, corresponding to an effective gravitational enhancement of $G_{\text{eff}}/G = 1 + 2\text{KAL}_0 \beta_c^2 \approx 1 + 2(5.521)(3.998)^2 \approx 177$. However, this linear estimate applies only far from the Vainshtein radius (see Section 6); inside r_V the non-linear kinetic term in (3) screens this amplification and restores GR.

4 Disformal Null Geodesic and Photon Deflection

4.1 Setup: photon propagation

Baryons and photons couple to the physical metric $g_{\mu\nu}$, not the disformal metric $\tilde{g}_{\mu\nu}$. A photon therefore follows the null geodesic of $g_{\mu\nu}$ in the weak-field perturbed Friedmann background:

$$ds^2 = -(1 + 2\Phi_N) dt^2 + (1 - 2\Phi_N) \delta_{ij} dx^i dx^j. \quad (13)$$

Standard GR light deflection by a point mass M gives the Einstein angle

$$\hat{\alpha}_{\text{GR}} = \frac{4GM}{c^2 b}, \quad (14)$$

where b is the impact parameter.

4.2 Modified gravitational potential

Dark matter in SSEE does not follow purely Newtonian gravity; it distributes in response to the combined force $F_{\text{tot}} = F_N + F_\phi$. For an isolated dark-matter-dominated system (galaxy cluster, compact group), the self-consistent equilibrium configuration produces an *effective* Newtonian potential Φ_{eff} that satisfies

$$\nabla^2 \Phi_{\text{eff}} = \frac{\rho_{\text{DM,eff}}}{2M_{\text{Pl}}^2}, \quad (15)$$

where $\rho_{\text{DM,eff}}$ is the dark matter density redistributed under the scalar force.

On scales $k < k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$, both the CDM sector ($\Omega_{\text{CDM}} = 0.160$) and the ϕ -DM sector ($\Omega_{\phi\text{DM}} = 0.160$) contribute, and the effective lensing mass fraction is $\Omega_{m,\text{eff}} = 0.320 \approx \Omega_{m,\text{CMB}}$ — identical to the standard ΛCDM value. On these scales the Einstein radius matches the GR prediction.

On scales $k > k_{\text{fs}}$, the ϕ -DM component free-streams out and only the CDM sector ($\Omega_{\text{CDM}} = 0.160$) remains. The effective gravitational source for photons is then suppressed by the factor $\mathcal{M} \approx 2$ relative to the ΛCDM expectation, and the Einstein radius is correspondingly reduced (see Section 7).

4.3 Total deflection angle outside Vainshtein radius

Outside the Vainshtein radius r_V (derived in Section 6), the scalar force operates at full strength. The total deflection of a photon passing an isolated dark-matter halo of effective gravitational mass M_{eff} at impact parameter b is, following the standard geodesic perturbation calculation (Will, 1993; Bartelmann & Schneider, 2001):

$$\hat{\alpha}_{\text{SSEE}} = \frac{4GM_{\text{eff}}}{c^2 b} \approx \beta_c \frac{4GM_{\text{bary}}}{c^2 b}, \quad (16)$$

where the second equality holds in the limit where the dark-matter distribution is entirely sourced by the scalar-force-enhanced coupling to the baryonic seed M_{bary} , and $\beta_c \approx 3.998$ is the disformal coupling.

Caveat: Equation (16) represents a working phenomenological estimate. A full solution requires self-consistent integration of the dark-matter distribution under both F_N and F_ϕ , which we defer to future N-body simulations. The result (16) should be understood as an upper bound on the lensing enhancement, applicable to systems dominated by scalar-force-driven dark matter accumulation.

5 Algebraic Emergence of $\mathcal{M}$

5.1 Einstein radius in SSEE

The Einstein ring is the locus where a source is perfectly aligned with the lens and observer. The Einstein radius θ_E satisfies the lens equation

$$\theta_E = \hat{\alpha}(\theta_E) \frac{D_{\text{LS}}}{D_S}, \quad (17)$$

where D_L , D_S , D_{LS} are angular diameter distances. Substituting (16) with $b = D_L \theta_E$:

$$\theta_E^2 = \beta_c \frac{4GM_{\text{bary}}}{c^2} \frac{D_{\text{LS}}}{D_L D_S}. \quad (18)$$

Comparing to the GR result $(\theta_E^{\text{GR}})^2 = 4GM_{\text{bary}} D_{\text{LS}}/(c^2 D_L D_S)$, we obtain

$$\boxed{\frac{\theta_E^{\text{SSEE}}}{\theta_E^{\text{GR}}} = \sqrt{\beta_c} = \sqrt{\mathcal{A}} \approx 1.9995.} \quad (19)$$

5.2 Algebraic identification with $\mathcal{M}$

The structural constant $\mathcal{M}$ is defined in equation (1):

$$\mathcal{M} = \frac{3\varphi + \pi}{4} = \frac{\beta_c}{2} \approx 1.9989. \quad (20)$$

Meanwhile, $\sqrt{\beta_c} = \sqrt{2\mathcal{M}}$. At the specific value $\mathcal{M} \approx 2$, we have $\sqrt{2\mathcal{M}} \approx \sqrt{4} = 2 \approx \mathcal{M}$, and more precisely:

$$\sqrt{\beta_c} = \sqrt{2\mathcal{M}} \approx 1.9995, \quad \mathcal{M} \approx 1.9989, \quad (21)$$

so $\sqrt{\beta_c}/\mathcal{M} = 1.00030$ (agreement to 0.03%).

This near-coincidence $\mathcal{M} \approx \sqrt{\beta_c}$ is a consequence of the structural postulate $\beta_c \equiv \mathcal{A} = 2\mathcal{M}$, combined with $\mathcal{M} \approx 2$. Concretely:

$$\mathcal{M} = 2 \iff \beta_c = 4 \iff \sqrt{\beta_c} = 2 = \mathcal{M}. \quad (22)$$

The algebraic values give $\mathcal{M} = 1.9989$ and $\beta_c = 3.9978$, both differing from the integer approximation by $< 0.06\%$.

Therefore, to within 0.03% , the lensing amplification factor coincides with the CMB horizon mapping factor:

$$\theta_E^{\text{SSEE}} \approx \mathcal{M} \times \theta_E^{\text{GR}}, \quad (23)$$

where the same $\mathcal{M}$ appears in:

- Paper 3: $r_{d,\text{eff}} = r_{d,\text{SSEE}}/\mathcal{M} \approx 147.2 \text{ Mpc}$
- Paper 6: $\Omega_{m,\text{CMB}} = \mathcal{M} \times \Omega_{m,\text{dyn}} = 0.3199$
- This paper: $\theta_E^{\text{SSEE}} \approx \mathcal{M} \times \theta_E^{\text{GR}}$

This triple appearance of $\mathcal{M}$ across the CMB, large-scale structure, and strong-lensing sectors is a structural prediction of the SSEE algebraic framework, not a tunable parameter.

6 Vainshtein Screening

6.1 Non-linear kinetic term

The X^2/M^4 term in the SSEE action (3) provides a Vainshtein-like screening mechanism analogous to Galileon and DBI models (Vainshtein, 1972; Deffayet et al., 2001; Babichev & Deffayet, 2013). When $X^2/M^4 \gg X/\text{KAL}_0$, the non-linear kinetic term dominates and the scalar degree of freedom becomes effectively massive around any overdensity, suppressing F_ϕ/F_N .

The crossover between the two kinetic regimes occurs at the gradient scale

$$|\nabla\phi|_* = \frac{M^4}{\text{KAL}_0}, \quad (24)$$

which defines the Vainshtein radius r_V via the gradient closure (11).

6.2 Vainshtein radius

Substituting the gradient closure $|\nabla\phi| = 2\text{KAL}_0\beta_c M_{\text{Pl}} |\nabla\Phi_N|$ and $|\nabla\Phi_N| \approx GM/(c^2 r^2) = r_S c^2/(2r^2)$ (where $r_S = 2GM/c^2$ is the Schwarzschild radius) into the crossover condition (24):

$$2\text{KAL}_0\beta_c M_{\text{Pl}} \frac{r_S c^2}{2r_V^2} = \frac{M^4}{\text{KAL}_0}. \quad (25)$$

Solving for r_V :

$$r_V = \left(\frac{\text{KAL}_0^2 \beta_c M_{\text{Pl}} r_S c^2}{M^4} \right)^{1/2}. \quad (26)$$

Equation (26) shows that r_V depends critically on the value of M , which is a free parameter in the IR EFT formulation of this paper but is fixed algebraically in Paper 10 (Almeida Vallejo, 2026d). Using $\text{KAL}_0 = 5.521$, $\beta_c = 3.998$, and $r_S = 2GM_{\text{obj}}/c^2$, we evaluate r_V in the two physically motivated limits:

IR limit ($M = M_{\text{Pl}}$, **reference value**): With $M = M_{\text{Pl}}$ the prefactor reduces to $\sqrt{\text{KAL}_0^2 \beta_c} \approx 11.04$, giving

$$r_V^\odot|_{M=M_{\text{Pl}}} \approx 11.04 \sqrt{(2953 \text{ m})(1.616 \times 10^{-35} \text{ m})} \approx 2.4 \text{ fm}, \quad (27)$$

$$r_V^{10^{12} M_\odot}|_{M=M_{\text{Pl}}} \approx 11.04 \sqrt{(2.95 \times 10^{15} \text{ m})(1.616 \times 10^{-35} \text{ m})} \approx 2.4 \text{ nm}. \quad (28)$$

Both values are far below any astrophysical scale of interest. At $M = M_{\text{Pl}}$ the non-linear kinetic term X^2/M^4 never activates on sub-Hubble scales, and Vainshtein screening is absent.

UV completion ($M = 8.81 \text{ meV}$, **Paper 10**): Paper 10 fixes $M^4 = 45\alpha^2 = 5\varphi^8$, giving $M \approx 8.81 \text{ meV}$. Since $r_V \propto M^{-2}$, the Vainshtein radius scales as $(M_{\text{Pl}}/M)^2 \approx 7.6 \times 10^{58}$ relative to the $M = M_{\text{Pl}}$ case:

$$r_V^\odot|_{M=8.81 \text{ meV}} \approx 1.8 \times 10^{44} \text{ m} \approx 4 \times 10^{17} H_0^{-1}, \quad (29)$$

$$r_V^{10^{12} M_\odot}|_{M=8.81 \text{ meV}} \approx 1.8 \times 10^{50} \text{ m} \approx 4 \times 10^{23} H_0^{-1}. \quad (30)$$

Both values vastly exceed the Hubble radius $H_0^{-1} \approx 4.4 \times 10^{26} \text{ m}$. Every astrophysical object therefore lies deep inside its own Vainshtein sphere, and the non-linear kinetic term dominates all local ($r \ll H_0^{-1}$) scalar dynamics.

Physical interpretation: Table 1 summarises the two regimes. The key distinction is between the *cosmological background* and *local perturbations*. The background kinetic energy satisfies $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$ (Paper 10), confirming that the Hubble flow is in the linear kinetic regime at both M values. Local scalar gradients around overdensities, however, are in the non-linear regime once $M \ll M_{\text{Pl}}$, exactly as in DGP and Galileon theories. This split — cosmological linear / local non-linear — is the hallmark of Vainshtein mechanisms in the literature (Babichev & Deffayet, 2013).

Table 1: Vainshtein radius and screening status for the two physically motivated values of M . The UV value follows from Paper 10 ($M^4 = 45\alpha^2$).

M value	$r_V^\odot$	$r_V^\odot/H_0^{-1}$	Screening active?
M_{Pl} (IR reference)	2.4 fm	$\sim 10^{-57}$	No
8.81 meV (UV, Paper 10)	$\sim 10^{44} \text{ m}$	$\sim 4 \times 10^{17}$	Yes — everywhere

When $M = 8.81 \text{ meV}$, every solar-system and galactic object is at $r \ll r_V$; the scalar-force ratio (Section 6) $F_\phi/F_N \propto (r/r_V)^{4/3} \rightarrow 0$ and GR is recovered with negligible fifth-force correction at all astrophysical scales.

6.3 GR compliance from selective coupling

Solar-system gravitational tests are nevertheless automatically satisfied by the *selective coupling* structure of the action (2):

- **Baryons** couple to the physical metric $g_{\mu\nu}$ only. They feel no direct force from the scalar gradient $\nabla\phi$.

- **Dark matter** couples to the disformal metric $\tilde{g}_{\mu\nu}$. It is the sole source of ϕ through the scalar EOM (9).
- In the solar system, the dark-matter density is $\rho_{\text{DM}} \approx 0.3 \text{ GeV}/\text{cm}^3$, far below galactic overdensities. The scalar field is barely sourced locally, so $\nabla\phi \approx 0$ and $\tilde{g}_{\mu\nu} \approx g_{\mu\nu}$.

Therefore all standard solar-system tests (Cassini time delay, perihelion precession, lunar laser ranging (Will, 2014)) pass at leading order without requiring any non-linear screening. This mechanism operates independently of the value of M and holds in both the IR ($M = M_{\text{Pl}}$) and UV ($M = 8.81 \text{ meV}$) limits.

6.4 Double GR protection at UV completion

With the UV value $M = 8.81 \text{ meV}$ established in Paper 10, the SSEE model benefits from *two independent mechanisms* that enforce GR compliance:

1. **Selective coupling (always active):** Baryons couple only to the physical metric $g_{\mu\nu}$ and feel no direct scalar force, regardless of M . This is the dominant mechanism in the IR limit.
2. **Vainshtein screening (active at UV completion):** With $M = 8.81 \text{ meV}$, every astrophysical object is at $r \ll r_V$ (Table 1). The scalar EOM inside r_V is governed by the non-linear term $\nabla \cdot [(|\nabla\phi|^2/M^4)\nabla\phi] = \beta_c \rho_{\text{DM}}/M_{\text{Pl}}$. For spherical symmetry this gives $\phi'(r) \propto r^{-2/3}$, and the dark-matter self-force satisfies

$$\frac{F_\phi}{F_N} \propto \left(\frac{r}{r_V}\right)^{4/3} \rightarrow 0 \quad (r \ll r_V), \quad (31)$$

suppressing the scalar contribution to dark-matter clustering at all sub-Hubble scales. Since $r/r_V \lesssim 10^{-17}$ for any solar-system measurement, the fifth-force correction is negligible at the level of current precision tests.

The two mechanisms are complementary: selective coupling protects baryonic matter unconditionally, while Vainshtein screening protects the dark-matter sector at UV completion. Importantly, the cosmological background Hubble flow remains in the linear kinetic regime ($\varepsilon = X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$, Paper 10), so the background equations of motion are unchanged by the UV correction. This is the standard separation between the cosmological linear background and the locally non-linear Vainshtein sector seen in DGP and Galileon theories (Babichev & Deffayet, 2013).

7 Strong Lensing Predictions

7.1 Two-sector lensing signature

The SSEE two-sector structure (Paper 6) generates a scale-dependent lensing signature. Define the lensing mass fraction as $f_{\text{lens}}(k) \equiv \rho_{\text{eff}}(k)/\rho_{\text{total}}$:

$$f_{\text{lens}}(k) = \begin{cases} 1 & k < k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1} \\ 1/\mathcal{M} \approx 0.500 & k > k_{\text{fs}} \end{cases}, \quad (32)$$

because on small scales only the CDM sector ($\Omega_{\text{CDM}} = 0.160$) contributes, while the ϕ -DM sector ($\Omega_{\phi\text{DM}} = 0.160$) is free-streaming. The ratio of CDM to total is $0.160/(2 \times 0.160) = 1/2 \approx 1/\mathcal{M}$.

This implies:

1. **Galaxy cluster scale** ($\sim \text{few Mpc}$, $k \lesssim k_{\text{fs}}$): Both sectors contribute. The Einstein radius matches the ΛCDM prediction for the same total matter content. No anomalous signal expected.
2. **Sub-Mpc scale** ($k \gtrsim k_{\text{fs}}$, **compact lenses**): The ϕ -DM sector free-streams. The effective lensing mass is $\sim \mathcal{M}$ times smaller than the inferred total mass. The Einstein radius is suppressed by $\sqrt{f_{\text{lens}}} = 1/\sqrt{\mathcal{M}} \approx 0.707$ relative to the ΛCDM expectation for the same total $\Omega_m = 0.320$.
3. **Comparison with baryonic mass**: For a system with measured baryonic mass M_{bary} and inferred total mass M_{total} , SSEE predicts $M_{\text{total}}/M_{\text{bary}} \lesssim \Omega_{m,\text{CMB}}/\Omega_b \approx 6.6$ on large scales, identical to ΛCDM .

7.2 Observable consequences in galaxy-scale lensing

For a galaxy of stellar mass $M_* \sim 10^{11} M_{\odot}$ at $z \sim 0.5$ lensing a source at $z \sim 2$, the typical Einstein radius in ΛCDM is $\theta_{\text{E}}^{\text{GR}} \sim 1''$. SSEE predicts the same value on large scales (both sectors present), but on sub-kpc scales the CDM-only sector reduces the effective lensing mass, potentially observable as:

- **Anomalous mass-to-light ratio**: The dynamical (velocity dispersion) mass $\sigma^2 r/G$ probes the local force F_{tot} , which is enhanced by the scalar force outside r_{V} . The lensing mass probes only Φ_N (metric perturbation). SSEE predicts a $\sim$ scale-dependent mass discrepancy between dynamical and lensing masses.
- **Radial Einstein ring profile**: The ratio $\theta_{\text{E}}(\theta)/\theta_{\text{E}}^{\Lambda\text{CDM}}(\theta)$ should show a transition near the angular scale corresponding to $\ell_{\text{fs}} \approx \pi/k_{\text{fs}} D(z) \sim 20'$ for a $z = 0.5$ lens.
- **Higher-order arcs**: Galaxy–galaxy lensing in JWST deep fields probes the projected mass on scales 1–10 kpc, where the two-sector suppression should appear as a systematic reduction in the shear signal relative to ΛCDM expectations.

7.3 Reference numbers

Table 2 summarises the key numerical predictions.

8 Falsifiable Predictions

We collect the pre-committed falsifiable predictions of this paper. A single decisive measurement inconsistent with any of these would falsify the corresponding sector of the SSEE framework.

1. **Free-streaming cutoff** ($k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$, Paper 6): DESI Year 3 and Euclid weak-lensing surveys will constrain the matter power spectrum at this scale to better than 5% by 2027. SSEE predicts a suppression of $\sim 30\%$ in $P(k)$ at $k > k_{\text{fs}}$ relative

Table 2: SSEE strong-lensing predictions compared to GR/ Λ CDM.

Observable	GR/ Λ CDM	SSEE	Scale
k_{fs} free-streaming cutoff	N/A	$0.493 h \text{ Mpc}^{-1}$	All
$\theta_{\text{E}}^{\text{SSEE}}/\theta_{\text{E}}^{\text{GR}}$ (large scale)	1	1 (both sectors)	$k < k_{\text{fs}}$
$\theta_{\text{E}}^{\text{SSEE}}/\theta_{\text{E}}^{\text{GR}}$ (small scale)	1	$1/\sqrt{\mathcal{M}} \approx 0.707$	$k > k_{\text{fs}}$
$M_{\text{dyn}}/M_{\text{lens}}$ ratio	1 (GR)	> 1 (fifth force on DM)	$r > r_{\text{V}}$
Vainshtein screening	N/A	$F_{\phi}/F_N \propto r^{4/3} \rightarrow 0$	$r \ll r_{\text{V}}$
GW speed anomaly (α_T)	0	0 (exact)	All

to Λ CDM (see Paper 6, Figure 2). *Falsification criterion:* A $< 5\%$ suppression at $k = 0.493 h \text{ Mpc}^{-1}$ would falsify the ϕ -DM mass $m_{\phi} = 5.71 \text{ eV}$.

- Lensing mass transition:** JWST strong-lensing of galaxy-scale systems at $z \sim 0.5$ should show a $\sim 30\%$ deficit in θ_{E} on scales $< 1 \text{ Mpc}$ relative to Λ CDM predictions using the total inferred $\Omega_m = 0.320$. *Falsification criterion:* If JWST finds θ_{E} consistent with Λ CDM on all scales $> 0.1 \text{ Mpc}$, the two-sector free-streaming prediction is falsified.
- Dynamical vs. lensing mass discrepancy:** Galaxy clusters at $z < 0.5$ should show $M_{\text{dyn}} > M_{\text{lens}}$ by a factor related to $\beta_c \approx 4$ outside r_{V} . *Falsification criterion:* If $M_{\text{dyn}}/M_{\text{lens}} = 1.00 \pm 0.05$ is confirmed across a sample of 100 clusters, the linear coupling picture is falsified.
- Gravitational wave speed:** $\alpha_T = 0$ (exact), same as GR. The GW170817 constraint $|\alpha_T| < 10^{-15}$ (LIGO/Virgo/Fermi, 2017) is already satisfied. *Falsification criterion:* Any detection of $\alpha_T \neq 0$ with current or future GW detectors would require modification of the SSEE gravitational sector.
- Scalar mass $m_{\phi} = 5.71 \text{ eV}$** (Paper 6): KATRIN neutrino mass measurement (2027) and PTOLEMY (2030) will probe m_{ϕ} in the eV range. If no species is found at this mass, the algebraic identification $m_{\phi} = (\Sigma m_{\nu}^{\text{active}}) \times H_0^{\text{alg}}$ is falsified.

9 Discussion

9.1 Consistency of MIRA across sectors

The appearance of $\mathcal{M} \approx 1.9989$ in three independent contexts — the CMB acoustic horizon (Paper 3), the two-sector mass ratio (Paper 6), and the lensing amplification (this paper) — is a structural prediction of the algebraic framework. It follows from the single postulate $\beta_c \equiv \mathcal{A} = (3\varphi + \pi)/2$:

$$\text{CMB: } r_{d,\text{eff}} = r_{d,\text{SSEE}}/\mathcal{M}, \quad (33)$$

$$\text{LSS: } \Omega_{m,\text{CMB}} = \mathcal{M} \Omega_{m,\text{dyn}}, \quad (34)$$

$$\text{Lensing: } \theta_{\text{E}}^{\text{SSEE}} \approx \mathcal{M} \times \theta_{\text{E}}^{\text{GR}}. \quad (35)$$

These three relations are not independent inputs; they follow from $\beta_c = 2\mathcal{M}$ combined with $\mathcal{M} \approx 2$. The algebraic near-coincidence $\mathcal{M} \approx \sqrt{\beta_c}$ (to 0.03%) is a consequence of $\mathcal{M} \approx 2$ and $\beta_c = 2\mathcal{M}$.

9.2 Limitations and open questions

The derivations in this paper adopt the quasi-static, non-relativistic limit and treat the scalar force in the linear regime (outside the Vainshtein radius). Several steps remain to complete the strong-gravity picture:

1. **Self-consistent DM distribution:** The estimate $\hat{\alpha}_{\text{SSEE}} \approx \beta_c \hat{\alpha}_{\text{GR}}$ assumes the dark-matter distribution is entirely determined by the scalar force. A proper calculation requires solving the coupled DM–scalar system numerically (N-body simulation with scalar force module).
2. **Full Vainshtein calculation:** The scaling relation $r_V \sim r_S^{1/2} \Lambda_{\text{UV}}^{-1/2}$ is parametric. Paper 10 now fixes $M = \varphi^2 5^{1/4} 1^{1/4} \approx 8.81 \text{ meV}$ algebraically, giving quantitative Vainshtein radii ($r_{V,\odot} \approx 1.84 \times 10^{44} \text{ m}$, Table 1); a full N-body calculation would convert the parametric scaling to a precise profile for individual halos.
3. **Relativistic corrections:** On cluster scales at $z \sim 0.5$, relativistic corrections to the geodesic ($\Phi_N \sim 10^{-4}$ – 10^{-3}) are small but not negligible at the precision of JWST.
4. **Baryonic feedback:** The lensing predictions in Table 2 are at the linear DM level. Baryonic feedback redistributes mass on sub-100 kpc scales and will modify the predicted Einstein ring profiles.

9.3 Connection to UV completion

The X^2/M^4 kinetic term in the SSEE action is the leading non-linear operator in a derivative expansion. Paper 10 of the SSEE series derives $M^4 = 45\alpha^2 = 5\varphi^8$ from the inflationary α -attractor parameter $\alpha = \varphi^4/3$, fixing

$$M = \varphi^2 5^{1/4} 1^{1/4} \approx 8.81 \text{ meV} \quad (36)$$

algebraically without free parameters. This is a *cosmological-scale* cutoff—thirty orders of magnitude below M_{Pl} —so no quantum-gravity UV completion is required. The EFT derivative expansion is valid over the entire cosmological regime; the X^2/M^4 operator is a dark-energy self-interaction, not a near-Planck correction.

10 Conclusions

We have derived the strong-gravity consequences of the SSEE-V3.6 disformal coupling and found:

1. **Gradient closure:** In the quasi-static limit, the scalar field profile tracks the Newtonian potential via $\nabla\phi = 2\text{KAL}_0\beta_c M_{\text{Pl}} \nabla\Phi_N$, with $\text{KAL}_0 \approx 5.521$ and $\beta_c \approx 3.998$ fixed algebraically.
2. **Lensing amplification:** The total deflection angle is amplified by β_c relative to the GR baryonic-mass prediction, yielding $\theta_{\text{E}}^{\text{SSEE}} \approx \sqrt{\beta_c} \approx \mathcal{M}$ times the GR Einstein radius.

3. **MIRA emergence:** The structural constant $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$ that maps the CMB acoustic horizon and the two-sector mass ratio also determines the lensing amplification to within 0.03% (via the near-coincidence $\sqrt{\beta_c} \approx \mathcal{M}$).
4. **GR compliance:** Paper 10 fixes $M \approx 8.81 \text{ meV}$, giving $r_V \sim (r_S/M^2)^{1/3} \approx 1.84 \times 10^{44} \text{ m}$ for the Sun—larger than the observable universe. Every astrophysical object sits deep inside its own Vainshtein sphere, where the scalar force is exponentially screened; GR tests are satisfied with double protection (screening plus the algebraic $\equiv 0$ from Paper 7).
5. **Two-sector signature:** SSEE predicts a scale-dependent lensing suppression near $k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$, testable with JWST and DESI.

Paper 8 thus extends the SSEE predictive reach from the linear cosmological regime into the non-linear strong-gravity regime, while maintaining the zero-free-parameter structure of the algebraic framework.

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